Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

3.6 Gas Pressure and Temperature

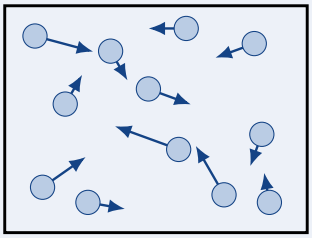
AQA Specification • §P3.3.1 Particle motion in gases • *P3.3 Particle model and pressure*

aP3-20 The molecules of a gas are in constant random motion. The temperature of the gas is related to the average kinetic energy of the molecules.
aP3-21 Changing the temperature of a gas, held at constant volume, changes the pressure exerted by the gas.
aP3-22 Students should be able to: explain how the motion of the molecules in a gas is related to both its temperature and its pressure; explain qualitatively the relation between the temperature of a gas and its pressure at constant volume.

Three ideas, each built on the one before: how the molecules move, how that makes a pressure, and what heating does to it.

1 · The molecules of a gas constant random motion

Key concept — motion and temperature



A longer arrow is a faster molecule.

The **molecules** of a gas are in **constant random motion**: they move all the time, in every direction, at a range of speeds. *Molecules* and *particles* mean the same thing here; the exam says molecules.

Some molecules are moving fast and some slowly, and they change speed every time they collide. So the gas has an **average** kinetic energy.

The temperature of a gas is related to the average kinetic energy of its molecules. Heat the gas and the molecules move faster *on average*, so the average kinetic energy rises, and that is what a higher temperature means.

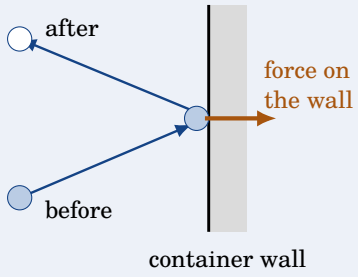
1) **State** two things about the motion of the molecules in a gas. **[2]**

2) **Compare** the motion of the molecules in two sealed containers of the same gas, one at 20 °C and the other at 80 °C. **[2]**

Check: your answer to question 2 has the word **average** in it. A few molecules in the cold gas are faster than a few in the hot gas; it is the *average* kinetic energy that goes with the temperature.

2 · Where gas pressure comes from — collisions with the walls

Key concept — pressure



A moving molecule that reaches the wall of its container **collides** with it and bounces back. Each collision pushes on the wall: a tiny **force**, for a tiny moment.

There are millions of millions of these collisions on every part of the wall every second, so the wall feels a **steady force**. Spread over the area of the wall, that force is the **pressure** of the gas.

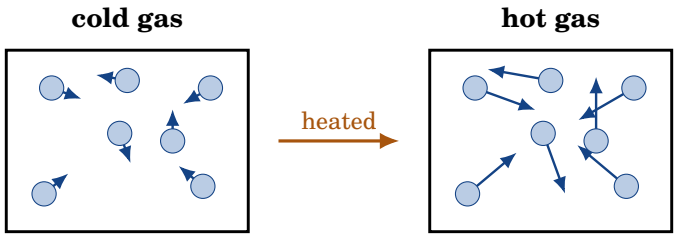
Say it in three links: collisions with the wall → a force on the wall → a pressure on the wall.

3) **Explain** how the molecules of a gas exert a pressure on the walls of their container. **[3]**

Check: your answer uses **collide**, **force** and **pressure**, in that order. Each word is caused by the one before it.

3 · Heating a gas at constant volume — faster molecules, more pressure

A **rigid, sealed** container keeps the volume of the gas the same and keeps every molecule inside. Heat it, and only the *motion* of the molecules changes.



Same container, same volume, same number of molecules. Only the arrows have changed.

Worked example — copy this shape

Explain why the pressure of a gas in a sealed, rigid container increases when the gas is heated.

1. Heating the gas gives its molecules more kinetic energy, so they move **faster on average**.
2. Faster molecules reach the walls sooner, so they collide with the walls **more often**.
3. Each collision is harder, so it exerts **a larger force** on the wall.
4. The volume is fixed, so the same wall area takes more collisions, each with more force: the **pressure increases**.

Check: a full answer has all four links, in order: faster → more often → more force → pressure up. To explain *cooling*, write the same four links with each one reversed.

4) **Explain** what happens to the pressure of the gas in a sealed, rigid container when the gas is **cooled**. Use the four links. **[4]**

4 · The same chain in the real world rigid containers, sealed

5) **Explain** why a car tyre's pressure gauge reads higher after a long, fast drive than it did before the car set off. Treat the tyre as a rigid, sealed container. **[3]**

6) **Explain** why the lid of a jar that was sealed while hot is harder to open once the jar has cooled in the fridge. The air outside the jar stays at the same pressure. **[3]**

Exam-style question [4 marks]

An aerosol can is a rigid, sealed metal container holding gas under pressure. Every can carries this hazard label: *"Pressurised container. Protect from sunlight. Do not expose to temperatures exceeding* 50 °C*."*

Use the particle model to explain why heating the can is dangerous.